



Vibration-Activated SMART CRO PROBE

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Here is a simple, low-cost, energy-saving and device-saving circuit for an electronics lab, service centre, electronics workshop, or wherever a CRO is in use. A vibration sensor attached to a CRO probe senses motion when you pick up the probe and turns on the CRO. It turns the CRO off when the probe is idle for a specified amount of time.

Generally, in a repair station, a CRO is used for a very short time. But in most cases, the user fails to switch off the CRO immediately after use. The service engineer mostly concentrates on faults rather than noticing whether the CRO is on or off. This results in wastage of power, reduction in CRO lifetime and leads to phosphorus burn in the cathode ray tube (CRT) due to electrons hitting the screen on the same spot for a very long time, which again leads to reduction of CRT lifetime. Fig. 1 shows the author's prototype.

Circuit and working

The block diagram of the vibration-activated smart CRO probe is shown in Fig. 2. A vibration sensor fixed on the CRO probe senses the vibrations picked up by the probe. This vibration signal is sensed, stored in a capacitor, compared with a reference voltage and fed to the relay driver to switch on the CRO. An additional circuitry is provided to switch off the CRO and give an audio acknowledgement in the form of a beeping sound.

The circuit diagram of the vibration-activated smart CRO probe is shown in Fig. 3. It is built around bridge rectifier DB107 (BR1), vibration sensor HDX-2NC connected

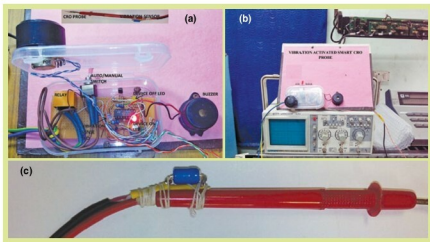


Fig. 1: Author's prototype: (a) internal part of the prototype, (b) prototype with CRO and (c) vibration sensor with CRO probe

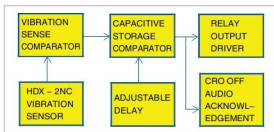


Fig. 2: Block diagram of the vibration-activated smart CRO probe

Test Points

Test point	Details
TP0	0V
TP1	Around 12V
TP2	Around 12V
TP3	1.2V
TP4	Around 12V after detecting vibration

across connector CON2, quad comparator LM324 (IC1), 1N4148 and 1N4007 diodes (D1 and D2), relay driver transistor BC547 (T1), 12V single-changeover relay (RL1) and a few other components.

As shown in Fig. 3, the circuit is built around IC LM324, which is a low-cost, quad operational amplifier (with four op-amps). Op-amps in this circuit are configured in comparator and inverted-buffer modes.

Here, HDX-2NC, a normally-closed vibration sensor is used. A vibration sensor is a small two-terminal component connected across connector CON2 in series with resistor R1. Point A in the circuit is connected to pin 3 of IC1, which is its non-inverting terminal. Under normal conditions, point A is connected to ground via sensor HDX-2NC. During vibration, its contact opens for a very small amount of time.

Op-amp A1 acts as an input comparator. Resistors R2 and R3 act as potential dividers and produce around 1.2V at point B in the circuit. When there is vibration, 12V is pro-